Comptia N+ Assignment

[N+ Network Fundamental and building networks]

OSI Layer

Task:

- **1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).**

  - **Aim**
    - To draw the OSI 7-layer model and identify each layer with its correct name and a real-world example of a device or protocol operating at that layer.

  - **Introduction**
    - The OSI (Open Systems Interconnection) Model is a reference model used to understand how data travels from one computer to another over a network. It divides network communication into seven layers. Each layer performs a specific function and works with the layers above and below it.

➢ Diagram of the OSI 7-Layer Model



- **Explanation of the Seven Layers**

Layer	Name	Function	Real-world example
7	Application	Provides network services to applications used by users.	HTTP, HTTPS, DNS
6	Presentation	Handles data formatting, encryption and compression.	JPEG, data encryption, TLS*
5	Session	Establishes, manages and terminates communication sessions.	NetBIOS, session management
4	Transport	Provides end-to-end data delivery and reliability when required.	TCP, UDP
3	Network	Handles logical addressing and routing of packets.	Router, IP
2	Data Link	Transfers frames within a local network using MAC addresses.	Network switch, Ethernet
1	Physical	Transmits raw bits through physical media or signals.	Hub, Ethernet cables, connectors

❖ **2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames. Write your answers as 'Scenario: Layer Name (Layer Number)'.**

➢ **Answer:-**
➢ a) Encrypting data before sending: **Presentation Layer (Layer 6)**
➢ b) Routing packets across networks: **Network Layer (Layer 3)**
➢ c) Converting data to electrical signals: **Physical Layer (Layer 1)**
➢ d) Opening a web page in your browser: **Application Layer (Layer 7)**
➢ e) Detecting and correcting errors in frames: **Data Link Layer (Layer 2)**

❖ **3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.**

➢ **Role of OSI Layers in WhatsApp**
➢ **Introduction**
- WhatsApp is a messaging application used to send and receive text messages, images, videos, voice messages, and documents over the internet. When a user sends a message through WhatsApp, different OSI layers work together to transfer the data from the sender's device to the receiver's device.

- **Working of OSI Layers in WhatsApp**
- The following OSI layers play an important role when sending or receiving a WhatsApp message.
- **1. Application Layer (Layer 7)**
  - The Application Layer provides network services to applications. When a user opens WhatsApp and sends a message to a friend, this layer allows the application to communicate over the network.
  - Example: Sending a text message or photo through WhatsApp.
- **2. Presentation Layer (Layer 6)**
  - The Presentation Layer is responsible for data formatting, encryption, and decryption in the conceptual OSI model. WhatsApp uses end-to-end encryption to protect personal messages and calls.
  - Example: Protecting a message so that only the intended participants can read it.
- **3. Transport Layer (Layer 4)**
  - The Transport Layer helps transfer data between devices. Protocols such as TCP provide reliable data delivery, while UDP is used by some real-time communication systems.
  - Example: Transporting application data between WhatsApp and its network services.
- **4. Network Layer (Layer 3)**
  - The Network Layer uses IP addresses and routing to move packets across different networks toward their destination.
  - Example: Routing data packets from the sender's phone across the internet toward the receiver's device.
- **5. Data Link Layer (Layer 2)**

- The Data Link Layer transfers frames across a local network, such as a Wi-Fi network, and uses MAC addresses for local delivery.
- Example: Transferring data from a phone through a Wi-Fi router.

➢ **Diagram: WhatsApp Message Transmission**



❖ **4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.**

➢ **Troubleshooting a Website Access Problem Using the OSI Model**

➢ **Scenario**

- A user is unable to access a website, but can successfully ping the website's IP address. We need to identify which OSI layer or layers might be causing the problem and explain the reason.

➢ **OSI Layers That Might Be Causing the Problem**

➢ **1. Application Layer (Layer 7)**

- The Application Layer is responsible for network services used by applications, such as web browsing through HTTP and HTTPS. If the web server's web service is unavailable or misconfigured, the website may not open even though the IP address responds to ping.
- Example: The web server is running, but its HTTP or HTTPS service is not working correctly.

➢ **2. Presentation Layer (Layer 6)**

- The Presentation Layer deals with data formatting and encryption in the conceptual OSI model. Problems with TLS/SSL configuration or certificate validation can prevent a secure website from opening.
- Example: The browser displays a security or certificate error when opening an HTTPS website.

➢ **3. Transport Layer (Layer 4)**

- The Transport Layer provides communication between applications using protocols such as TCP. A website may fail to load if the required TCP port is blocked or the web server is not accepting connections.
- Example: A firewall blocks TCP port 443, which is commonly used for HTTPS, or port 80, which is commonly used for HTTP.

➢ **4. Application Name Resolution (DNS)**

- If the website is accessed using its domain name, a DNS problem may prevent the browser from finding the correct IP address. Even if pinging the IP address works, accessing the domain name may fail.
- Example: The browser cannot resolve www.example.com to its IP address because of an incorrect DNS configuration.

➢ **Reasoning**

- A successful ping to the website's IP address generally indicates that the destination is reachable through the network using ICMP. However, ping does not verify that the website's HTTP or HTTPS service is working.
- Therefore, the problem may be related to the Transport Layer (Layer 4), Application Layer (Layer 7), TLS configuration, or DNS resolution. The Network Layer is less likely to be the main cause if the correct destination IP responds to ping, although some network problems may still affect web traffic specifically.